RAG Architecture and Core Components

A typical RAG system consists of several key components working together seamlessly. The first component is the knowledge base or corpus, which contains documents, articles, or data that the system can retrieve from. This corpus is preprocessed and stored in a vector database or semantic search engine for efficient retrieval. The retriever component uses dense vector embeddings to perform similarity search, finding documents most relevant to the user's query. Modern retrievers employ embedding models like BERT, MPNet, or specialized domain models that encode documents and queries into high-dimensional vector spaces. The ranking component orders retrieved documents by relevance, often using multiple ranking strategies to ensure the most pertinent information appears first. The context preparation stage formats retrieved documents into a prompt suitable for the generative model. The generator component is typically a large language model that takes the user query along with retrieved context and produces the final response. The entire pipeline operates in real-time, enabling interactive applications. Advanced RAG systems implement reranking stages, multi-hop retrieval for complex queries, and caching mechanisms for improved performance.